Ertel Potential Vorticity Computation on Pressure Levels

PV_Ertel_Computation Project
`https://github.com/yanxingjianken/PV_Ertel_Computation`

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1 Ertel PV in Height Coordinates

Ertel's potential vorticity [1] in height coordinates (x, y, z) is:

$$\mathrm{EPV} = \frac{1}{\rho}\left(\vec{\omega}_a \cdot \nabla\theta\right) \tag{1}$$

where:

- ρ = density $[\mathrm{kg\,m^{-3}}]$
- $\vec{\omega}_a = \nabla \times \vec{u} + 2\vec{\Omega}$ = absolute vorticity $[\mathrm{s^{-1}}]$
- $\theta = T(p_0/p)^{R_d/c_p}$ = potential temperature [K]

In component form:

$$\vec{\omega}_a = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z},\ \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x},\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f\right) \tag{2}$$

$$\mathrm{EPV} = \frac{1}{\rho}\left[\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)\frac{\partial \theta}{\partial x} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)\frac{\partial \theta}{\partial y} + (\zeta + f)\,\frac{\partial \theta}{\partial z}\right] \tag{3}$$

where $\zeta = \partial v/\partial x - \partial u/\partial y$ is the relative vorticity and $f = 2\Omega \sin\phi$ is the Coriolis parameter.

The MPAS-Atmosphere model [4] computes EPV on its native height-based, terrain-following vertical coordinate using finite-volume flux divergence on a Voronoi mesh (see `mpas_pv_diagnostics.F`).

2 Isobaric Transformation

Under the hydrostatic approximation $\partial p/\partial z = -\rho g$, the vertical coordinate transforms from z to p:

$$\frac{\partial}{\partial z} \to -\rho g \frac{\partial}{\partial p} \tag{4}$$

Applying this to Eq. (1):

$$\mathrm{EPV} = \frac{1}{\rho}\left[\frac{\partial w}{\partial y}\frac{\partial \theta}{\partial x} - \frac{\partial w}{\partial x}\frac{\partial \theta}{\partial y} + \rho g\left(\frac{\partial v}{\partial p}\frac{\partial \theta}{\partial x} - \frac{\partial u}{\partial p}\frac{\partial \theta}{\partial y} - (\zeta + f)\frac{\partial \theta}{\partial p}\right)\right] \tag{5}$$

Neglecting terms involving the horizontal gradient of vertical velocity w (valid at synoptic and larger scales), we obtain the **isobaric Ertel PV**:

$$\boxed{\mathrm{PV} = -g\left[\frac{\partial v}{\partial p}\frac{\partial \theta}{\partial x} - \frac{\partial u}{\partial p}\frac{\partial \theta}{\partial y} + (f + \zeta)\frac{\partial \theta}{\partial p}\right]} \tag{6}$$

In PVU ($1\,\mathrm{PVU} = 10^{-6}\,\mathrm{K\,m^2\,kg^{-1}\,s^{-1}}$):

$$\mathrm{PV_{PVU}} = \mathrm{PV_{SI}} \times 10^6 \tag{7}$$

3 Full 3-Term vs Simplified Formula

Eq. (6) contains three terms:

1. **Vortex stretching**: $(f+\zeta)\,\partial\theta/\partial p$
2. **Horizontal shear (1)**: $(\partial v/\partial p)(\partial\theta/\partial x)$
3. **Horizontal shear (2)**: $-(\partial u/\partial p)(\partial\theta/\partial y)$

The simplified (ERA5-like) formula retains only term (1):

$$\mathrm{PV_{simple}} = -g\,(f+\zeta)\,\frac{\partial\theta}{\partial p} \tag{8}$$

This approximation is valid when the horizontal shear of the vertical wind shear is small compared to planetary vorticity stretching — true for most synoptic-scale flows but not near strong fronts or jets.

In practice, Eq. (8) is what ERA5 reports as potential vorticity on pressure levels. Our validation against ERA5 native PV shows that the simplified formula matches better (RMSE ~ 0.21 PVU at 500 hPa) than the full 3-term formula (RMSE ~ 0.35 PVU), because the additional shear terms introduce numerical noise that outweighs their physical contribution at the resolutions tested.

4 Moisture Correction

In the presence of water vapor, the virtual temperature T_v replaces T:

$$T_v = T\,(1 + 0.61\,q) \tag{9}$$

where q is specific humidity [$\mathrm{kg\,kg^{-1}}$]. The virtual potential temperature is then:

$$\theta_v = T_v\left(\frac{p_0}{p}\right)^{R_d/c_p} \tag{10}$$

Using θ_v instead of θ in Eq. (6) provides a moisture correction that is largest in the tropical boundary layer (where $q \sim 0.015$–$0.020\,\mathrm{kg\,kg^{-1}}$, giving $T_v - T \sim 3$–$4\,\mathrm{K}$). At mid-tropospheric levels and above, $q \ll 0.001$ and the correction is negligible.

Our ERA5 validation shows that the moisture correction reduces RMSE by ∼0.001 PVU at 500 hPa — a negligible improvement for synoptic-scale studies, but potentially relevant for tropical or boundary-layer applications.

5 Numerical Discretization

5.1 Vertical Derivatives

Following MPAS `calc_vertDeriv_center` and `calc_vertDeriv_one`, we use:

- **Interior** ($k = 1, \ldots, N-2$): centred difference

$$\left.\frac{\partial f}{\partial p}\right|_k = \frac{1}{2}\left(\frac{f_{k+1}-f_k}{p_{k+1}-p_k} + \frac{f_k - f_{k-1}}{p_k - p_{k-1}}\right) \tag{11}$$

- **Top** ($k = N - 1$, lowest pressure): one-sided backward

$$\left.\frac{\partial f}{\partial p}\right|_{N-1} = \frac{f_{N-1} - f_{N-2}}{p_{N-1} - p_{N-2}} \tag{12}$$

- **Surface** ($k = 0$, highest pressure): one-sided forward

$$\left.\frac{\partial f}{\partial p}\right|_{0} = \frac{f_1 - f_0}{p_1 - p_0} \tag{13}$$

Pressure levels are ordered surface → top of atmosphere (highest to lowest pressure) throughout.

5.2 Horizontal Derivatives

On a regular latitude–longitude grid:

$$\frac{\partial f}{\partial y} = \frac{1}{R_E}\frac{\partial f}{\partial \phi} \approx \frac{1}{R_E}\frac{\Delta f}{\Delta \phi} \tag{14}$$

$$\frac{\partial f}{\partial x} = \frac{1}{R_E \cos\phi}\frac{\partial f}{\partial \lambda} \approx \frac{1}{R_E \cos\phi}\frac{\Delta f}{\Delta \lambda} \tag{15}$$

where $R_E = 6.371 \times 10^6\,\mathrm{m}$ and $\Delta\phi$, $\Delta\lambda$ are computed via `numpy.gradient`.

5.3 Pole Handling

At the poles ($\phi = \pm 90°$), $\cos\phi = 0$ and $dx \to 0$, leading to a singularity in $\partial/\partial x$. We clip $\cos\phi$ to a minimum value corresponding to half a grid spacing from the pole:

$$\cos\phi_{\min} = \cos(90° - 0.5\,\Delta\phi) \tag{16}$$

For a 1° grid, $\cos\phi_{\min} \approx 0.0087$.

6 Hybrid Sigma-Pressure Levels

For models using a hybrid vertical coordinate (e.g., CESM2/CAM6), the pressure at model level k is:

$$p_k = a_k\, p_0 + b_k\, p_s \tag{17}$$

where a_k ("hyam") and b_k ("hybm") are the hybrid coefficients at model midpoints, $p_0 = 100{,}000\,\mathrm{Pa}$ is the reference pressure, and p_s is the surface pressure.

When a_k and b_k are available, the per-column actual pressure $p_k(i, j)$ replaces the 1-D approximate pressure level array in all vertical derivative computations (Sec. 5.1). This is critical for accurate PV near the surface, where hybrid levels deviate most from pure pressure levels.

Note for CESM2-LENS2 users: The static grid file on AWS S3 (`atm/static/grid.zarr`) contains NaN values for the hybrid coefficients as of 2026-06-01. Users must obtain a_k and b_k from alternative sources (e.g., CAM initial condition files, CESM source code) to use this feature.

7 Validation Methodology

7.1 ERA5 Comparison

We validate against ERA5 native potential vorticity on pressure levels (product: `reanalysis-era5-pressure-le` variable: `potential_vorticity`).

For a test case (2025-01-08 00Z, all 37 pressure levels, global 0.25°):

Level	RMSE (simple, PVU)	Bias (PVU)	Correlation
850 hPa	3.53	+0.10	0.28
500 hPa	0.21	+0.01	0.96
250 hPa	1.09	-0.03	0.97

The poor 850 hPa comparison is expected: pressure-level PV products are least accurate near the surface where terrain-following coordinates differ most from pressure coordinates. The excellent 500 hPa and 250 hPa comparisons confirm the validity of the isobaric PV computation for the free troposphere and lower stratosphere.

7.2 CESM2-LENS2 Quick-Look

Applied to member `r10i1181p1f1` (2010-01-01), with 32 hybrid levels using approximate pressure. PV values are physically reasonable: -1.8 to $+3.1$ PVU at $\sim$525 hPa, and -10.3 to $+11.8$ PVU at $\sim$233 hPa.

References

References

[1] Ertel, H. (1942). Ein neuer hydrodynamischer Wirbelsatz. *Meteorologische Zeitschrift*, 59, 277–281.

[2] Holton, J. R. & Hakim, G. J. (2013). *An Introduction to Dynamic Meteorology*, 5th ed. Academic Press.

[3] Bluestein, H. B. (1992). *Synoptic-Dynamic Meteorology in Midlatitudes*, Vol. I. Oxford University Press.

[4] Skamarock, W. C. et al. (2012). A description of the Advanced Research WRF version 3. NCAR Technical Note NCAR/TN-475+STR.

[5] MPAS-Atmosphere source code. `mpas_toolchain/mpas/src/core_atmosphere/diagnostics/mpas_pv_di`

[6] Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146, 1999–2049.

[7] Rodgers, K. B. et al. (2021). The CESM2 Large Ensemble. *Earth System Dynamics*, in preparation. Data: `https://registry.opendata.aws/ncar-cesm2-lens/`